

Multilayered Magnetic Recording Media Enabled by a $\text{Pt}_{1-x}\text{Ta}_x$ Intermediate Layer

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Multilayered magnetic recording media using a Pt based interlayer leads to the formation of unwanted precipitates and degrades the magnetic properties of surrounding L1₀-FePt. We show that Ta addition to a Pt based interlayer is an effective method to structurally, chemically, and magnetically stabilize multilayered magnetic recording structures of the form FePt / Pt / FePt. Tantalum supersaturates the Pt crystal structure forming a metastable FCC structure at the nanoscale which can support L1₀-FePt within the first and second magnetic recording layers.

Index Terms— Heat-assisted magnetic recording (HAMR), multi-level magnetic recording, interlayer, intermediate layer.

I. INTRODUCTION

MULTILEVEL magnetic recording is a promising method that can achieve significantly improved data storage capabilities over conventional binary technologies. In particular, for heat assisted magnetic recording, a zero magnetization level state can be added by writing in zero recording field. Although it has been shown that a third zero polarization state can improve linear storage densities by nearly 60%, development of a suitable multilayered magnetic recording media has remained difficult [1, 2, 3].

The structure of multilevel magnetic recording media consists of two magnetic recording layers (MRL) separated by a non-magnetic interlayer material. The structure is of the form MRL-A / interlayer / MRL-B where either MRL consists of L1₀-FePt and a grain boundary segregant. Currently, there exists few suitable interlayer materials that integrate well with adjacent MRLs. Design requirements suggest that the interlayer should support good (001) texture of the second FePt layer and

maintain immiscibility with FePt and common grain boundary materials, e.g. *a*-C and *h*-BN, at temperatures near 650 °C.

Platinum is one interlayer material layer that has received interest due to its small lattice mismatch of ~1.3% with L1₀-FePt [3]. However, Pt tends to intermix with the surrounding FePt modifying the stoichiometry of the adjacent FePt phase or forming an ordered FePt₃ phase. In general, the FePt / Pt / FePt system equilibrates toward single phase FePt; the FePt / FePt₃ / FePt system only equilibrates at roughly 46 vol% FePt₃.

The complex phase equilibria of the FePt / Pt / FePt system makes it especially interesting to study regarding spacer layer interdiffusion. In particular, certain additives could be used to prevent interdiffusion and promote immiscibility between the interlayer and surrounding MRLs. One material that is commonly used as a diffusion barrier is Ta. Tantalum exhibits good solubility within FCC Pt and immiscibility with Fe at high temperatures. Noting the appealing properties of Ta, we have performed an experimental survey evaluating the role of Ta addition to Pt interlayers as a means to inhibit interdiffusion.

II. METHODS

Continuous films of FePt (5nm) / Pt_{1-x}Ta_x (3nm) / FePt (5nm) were prepared by magnetron sputtering onto sputtered MgO (002) underlayers prepared on Si substrates at T = 725°C. The Pt_{1-x}Ta_x layer was prepared by co-sputtering from pure Ta and Pt targets of 3N purity. The Ta content was modified by adjusting the power densities, roughly 0.75 – 2 w/cm², of either Pt or Ta target.

Granular FePt / *h*BN media was prepared onto single crystal MgO (002) substrates at T = 750°C following the process outlined by C. Xu et al. [4]. Media of the approximate structure FePt-*h*BN (4.5 nm) / Pt₂Ta-*h*BN (2nm) / FePt-*h*BN (4.5 nm) are presented below.

III. RESULTS AND DISCUSSION

X-ray diffraction (XRD) data are shown in Fig. 1 for continuous tri-layers of FePt / Pt / FePt deposited onto sputtered MgO (002) underlayers grown on Si substrates. Without inclusion of a Ta additive the Pt intermixes with the adjacent FePt layers forming a stable FePt₃ layer, see Fig. 1 (a). The Pt interlayer consumes the Fe contained within the FePt phase to form FePt₃

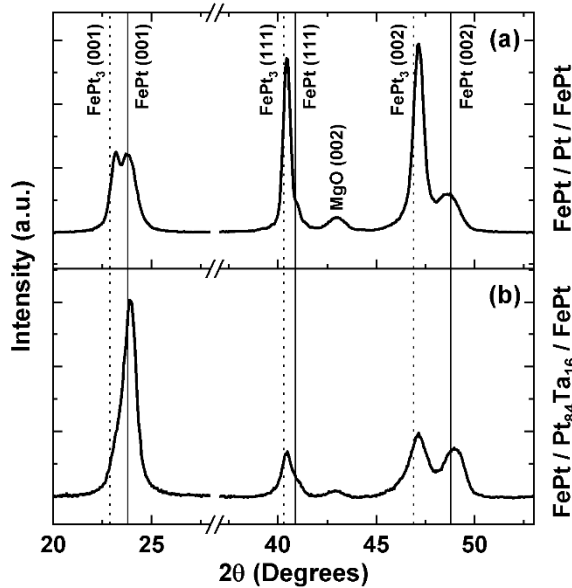


Fig. 1: XRD patterns showing continuous tri-layers of FePt / Pt / FePt (a) and FePt / Pt₈₄Ta₁₆ / FePt (b) grown on Si / Ta / Cr / MgO underlayers. Dashed reference lines represent FePt₃ and solid reference lines represent FePt peak positions.

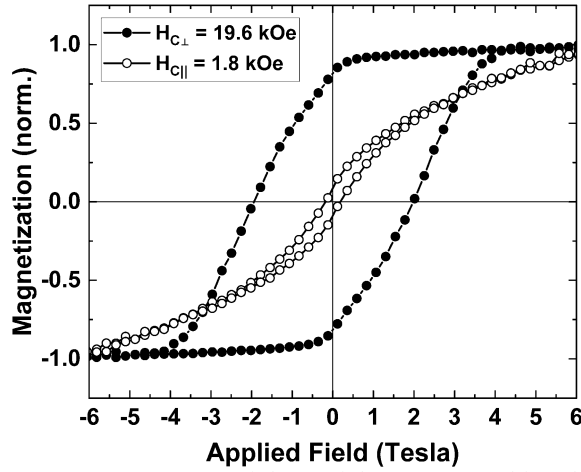


Fig. 2: Room temperature magnetic hysteresis loops as measured from the out-of-plane direction (filled circles) and in-plane direction (open circles). Data is provided for multilayer magnetic recording media of the general structure FePt-hBN (4.5 nm) / Pt₂Ta-hBN (2nm) / FePt-hBN (4.5 nm).

precipitates. Available Pt from the interlayer and the loss of Fe due to the formation of FePt₃ cause an Fe deficiency within the L1₀-FePt phase. This has been linked to degraded magnetic properties in similar systems [5].

Introducing an additive like Ta stabilizes the tri-layers by inhibiting interdiffusion. **Fig. 1 (b)** shows the XRD data for a continuous FePt / Pt₈₄Ta₁₆ / FePt tri-layer. All intensities of peaks associated with the FePt₃ phase were reduced. Additionally, the ordering of the L1₀-FePt phase was enhanced significantly from $I_{(001)}/I_{(002)} = 1.8$ to 2.5 for the Pt and Pt₈₄Ta₁₆ interlayers respectively. This suggests that Ta doped interlayers promote improved FePt stoichiometry compared to pure Pt interlayers.

Additional multilayered magnetic recording media was fabricated using a Pt-Ta interlayer to evaluate the role of Ta addition within granular media. The media was fabricated on single crystal MgO (002) substrates. The room temperature magnetic hysteresis data for media of the general structure FePt-hBN (4.5 nm) / Pt₂Ta-hBN (2nm) / FePt-hBN (4.5 nm) are provided in **Fig. 2**. The out-of-plane and in-plane coercivities were measured to be 19.6 and 1.8 kOe respectively. No step was observed along the out-of-plane direction at low applied field like other FePt / Pt / FePt systems [3, 6]. The out-of-plane hysteresis of this system was smooth and continuous.

The absence of a step within the magnetic hysteresis is likely attributed to exchange decoupled MRLs exhibiting comparable magnetic properties, i.e. MRL-A and MRL-B have similar anisotropy constants. Further evaluation of available characterization techniques is likely required to confirm such exchange decoupling within magnetically hard multilayered recording systems like this.

One method that has provided some insight regarding exchange decoupled and magnetically hard multilayers has been flip-disk bonding. Using this method we have fabricated multilayered recording media with an interlayer thickness of ~50nm. At this thickness it is reasonable to assume that there is no magnetic exchange coupling between either MRL. The character of the out-of-plane hysteresis provided in **Fig. 2** aligns well with what is observed for these bonded structures.

Fig. 3 provides a cross-sectional STEM-HAADF micrograph of the same FePt-hBN (4.5 nm) / Pt₂Ta-hBN (2nm) / FePt-hBN(4.5 nm) granular media structure. The micrograph shows

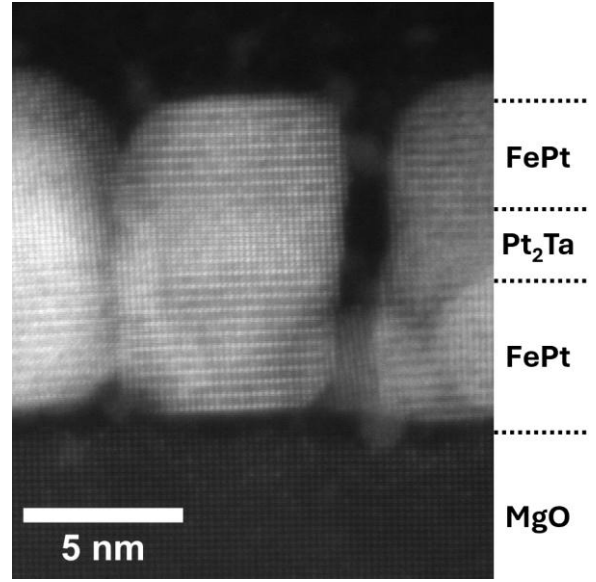


Fig. 3: Cross-sectional scanning transmission electron micrograph of multilayered magnetic recording media with the general structure FePt-hBN (4.5 nm) / Pt₂Ta-hBN (2nm) / FePt-hBN (4.5 nm). Data was collected using a high-angle annular dark-field detector and imaged along [001] MgO substrate.

a well isolated multilayered grain with dimensions 6.7×10.0 nm. The L1₀ ordering of FePt is easily observed within the first and second magnetic recording layers. The intermediate layer, Pt₂Ta, is homogeneous and does not show an FePt₃ phase. Notably, the Pt₂Ta does not crystallize to the known orthorhombic structure, rather, it retains the FCC structure associated with Pt. This was also confirmed via XRD suggesting that Pt₂Ta forms a metastable FCC structure at small dimensions.

IV. CONCLUSION

Multilayered magnetic recording media of the general structure FePt-hBN (4.5 nm) / Pt₂Ta-hBN (2nm) / FePt-hBN (4.5 nm) has been demonstrated. The addition of Ta to the interlayer prevents the formation of unwanted FePt₃ precipitates by inhibiting interdiffusion within the multilayered structure. It is shown that Pt can accommodate a large amount of Ta forming a metastable FCC structure at 33 at% Ta. This promotes L1₀ ordering within the first and second magnetic recording layers which can enable multistate magnetic recording.

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